MEAM 210: Statics and Strength of Materials (Fall 2012) (updated 9/26/12, changes marked in red)

Course: MEAM 210: Statics and Strength of Materials

Time and Location:

Lecture:	MWF 10:00-11:00 AM	Moore 216
Recitation (201):	R 10:30-11:30 AM	Towne 309
Recitation (202):	R 12:00-1:00 PM	Towne 321
Recitation (203):	R 1:30-2:30 PM	Towne 315

Credit: 1 CU

Instructor: Professor Kevin T. Turner
office: 245 Towne
e-mail: kturner@seas.upenn.edu
office hours: Monday 3-4 PM, Friday 3-4 PM

Teaching Assistants: Nathan Ip, ipnathan@seas.upenn.edu, recitation 202
office hours: Mon 4-5 PM (in 231 Fisher-Bennett Hall)

Helen Minsky, hminsky@seas.upenn.edu, recitations 201 and 203
office hours: Mon 5-6 PM (in 231 Fisher-Bennett Hall)

Textbooks:
- Required Textbook: R.C. Hibbeler, Statics and Mechanics of Materials, 3rd ed. (2010), ISBN-13 978-0132166744
- Lecture note handouts posted on course website
- Supplementary Books: on reserve at the library, see course website for details

Website: **https://courseweb.library.upenn.edu/**

Please check the course website regularly! On it you will find:
- copies of lecture notes from the course
- copies of required readings
- copies of and links to supplementary materials
- announcements regarding homework assignments

Course Description: This course is designed for sophomore-level engineering students who have completed MEAM 110 & 147 (or Physics 150/AP Physics). MEAM 210 is based on Newtonian physics and focuses on problems in structural mechanics and materials. The purpose of the class is to teach you basic concepts and methods in analyzing how loads (i.e. forces and moments) are distributed in structures and how materials respond to these mechanical loads. This course is fundamental to many engineering courses, and introduces strategies for problem solving. Basic physical principles are explored to understand mechanical phenomena and to design safe structures and machines that meet desired functions.

Course Format: This course meets for three lectures and one recitation section per week. Lectures will include a discussion of essential concepts, applications, and relevant example problems. Questions and discussion are strongly encouraged in lecture! Recitation sections will focus on additional example problems and student questions.

Prerequisites: MEAM 110/147 or Physics 150/AP Physics

Assessment/Grading:

Homework	20 %
Project	5 %
Two in-class exams	40 %
Final exam	35 %

Policies and Procedures:

HW Assignments:

Homework problems will be assigned regularly and are **due at 10 AM in lecture** on the dates indicated on the schedule. **Homework sets must be handed in before lecture begins**. **HW not handed in on time will be considered late and will be given a grade of 0.** This policy is firm and will be enforced uniformly; exceptions will only be made for medical reasons (in which case, a doctor's note is required) or a serious personal emergency (in which case, a note from your academic advisor is needed). HW can be handed in early in lecture, to your TA in recitation, or in office hours to me or the TAs. Do not leave HW in Prof. Turner's mailbox or the TAs mailboxes. If you have trouble getting to lecture on time, consider handing your HW in early. If you have a serious problem that will prevent you from handing in HW on time, please contact Prof. Turner (not your TA) **before the due date**.

The HW sets are designed to reinforce the material covered in lecture and to prepare you for the exams – spending time on the HWs is essential to doing well in this course as it allows you to learn the concepts and develop the strategies needed for solving problems on the exams!

Each homework problem must start at the top of a new page (front or back). The following information must be included: (1) your name and recitation section number, (2) problem number, (3) all appropriate diagrams (including a free-body diagram) (4) any assumptions, and (5) a step-by-step solution for each problem. Please refer to the example HW solution that is posted on the course website.

To receive full credit, your solution must include appropriate units and be expressed with the appropriate number of significant digits. HW should be neatly prepared and easy to follow - illegible HW will receive a grade of 0. The lowest score of all your HW assignments will be dropped.

Conduct:

The instructor, the Department of Mechanical Engineering and Applied Mechanics, and the University of Pennsylvania expect the highest standards of honesty and integrity in the academic performance of its students. Any student caught cheating in this course will receive an F and the case will be reported to the Dean's Office.

Collaboration with your classmates is important and can be very helpful to learning, but it is important to understand the allowable level of collaboration. You are encouraged to discuss homework and course material with the instructor, teaching assistants, and classmates. However, your submitted homework assignments and exams must involve only your own effort. Be careful to not rely too much on help from classmates when completing your HW assignments as there are two likely consequences: (1) You will do poorly on the exams when you are completely on your own. (2) If you overstep the bounds of "getting help" to "copying", you will likely be caught and penalized for cheating.

There are many resources available to students to help with learning, including web-based resources, additional books on reserve at the library, the instructor, and the TAs – please make extensive use of these resources. In the past, students have also located copies of solutions to the HW problems in the book. These HW solutions are NOT an acceptable resource. Using a solutions manual or copies of HW solutions to help you complete the HW is cheating and is unfair to your fellow classmates that do not have access to the solutions.

We have seen copying and cheating before and we know how to catch it. Please don't do it, because it is not worth it.

Quizzes:

There will be no quizzes in this course.

Project:

There will be a three-part mini-design project assigned in this class. This project will be team based and will require you to apply concepts learned in class to analyze a real structure. Details will be provided at the time the project is assigned.

Exams:

Two in-class exams will be held on the dates and times listed on the schedule. The exams will be 50 minutes. Calculators are allowed. The exam will be closed notes, closed book, but one 8.5" x 11" handwritten formula sheet (both sides) is allowed. Students who miss an exam will receive a zero – except in cases of very extenuating circumstances (acceptable proof of these circumstances must be provided by the student).

Grade Questions:

All grade-related questions must be submitted in writing (via e-mail) within five calendar days of the regular return date of the relevant assignment/exam. Please note, that additional points might be subtracted if the requested regrade reveals additional student errors.

Disability Accommodation:

If you have a physical or learning disability that the instructor should be aware of, please arrange for an individual consultation with the instructor to discuss accommodation.

Advice:

Expectations:

It is expected that you will attend all lectures and recitations. There is material and insight that is provided in lecture and recitation that may not be in the textbook or the posted lecture notes. It is also expected that you will read the sections of the textbook noted on the course schedule before lecture. The lectures are designed assuming that you have read the relevant sections of the textbook and the lectures will make much more sense if you have done the reading. There is a good chance that you will eventually read the textbook sections when you are completing the HW, so why not read it beforehand and be well prepared for lecture? By reading the textbook beforehand, you will be ready to ask questions and participate in the discussions in lecture.

Lecture Note Handouts:

Lectures will be taught from a combination of slides and board work. A copy of the lecture notes will be posted on the course website by 6:00 pm on the day before lecture. You are strongly encouraged to print a copy of these lecture notes and bring them to lecture. Having a printed version of the lecture notes will allow you to more fully listen and participate in the discussion in lecture and allow you to focus your note taking in class on the material that is presented on the board.

These lecture notes are a resource and are not a replacement for attending lecture, participating in recitation, or reading the textbook. These lecture notes will contain essential information and guide the lectures, but it is entirely possible that an exam or HW problem may be based on something that was only covered on the board in lecture or only covered in the book. If you miss a lecture, please be sure to get notes from one of your classmates.

Some keys to getting an A in this course:

1) Attend all classes, pay attention, and participate.
2) Read the relevant sections of the text before attending lecture.
3) Ask questions in class. Questions allow the instructor to understand what concepts are not clear and your classmates will appreciate your questions as they likely have the same questions.
4) Work with your classmates. Working together allows you to challenge each other, learn from each other, and check that you really know what you think you know.
5) Think for yourself. When working with others, be sure that your solutions to the problems are your own work and that you can reason through every step of the problems on your own.
6) Come to office hours if you are stuck. Ask well-prepared questions, not just about specific problems, but also about the concepts you are learning.
7) Complete all HW assignments on time or ahead of time. Do not leave your homework until the night before. Do extra problems from the textbook or supplementary texts.
8) Follow up. Check the HW and exam solutions. If they do not make sense, ask your classmates, your TA, or your instructor about it in class, recitation, or office hours. If you do poorly on a midterm, see the instructor or your TA to discuss it in order to find out what's going wrong.

IF YOU ARE HAVING TROUBLE IN THE COURSE, TALK TO THE INSTRUCTOR AS SOON AS POSSIBLE.